Principles of Quantum Mechanics

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1 Quantum state

Quantum mechanics is the fundamental theory that describes physical reality. The philosophy of science dictates that any scientific theory must account for any observations in the real world. Despite the triumph of classical physics for more than two centuries, it eventually disagreed with experimental observation, two of the most important ones are the photoelectric effect and double-slit experiment. Two types of distinctively different objects in classical physics are particles and waves, however, the two experiments above demonstrates the fundamental objects of material world behave as both, thus exhibits what is known as the **wave-particle duality**. The result of experiments hint at a more fundamental and intricate theory of reality, classical mechanics was therefore replaced by quantum mechanics. However, classical mechanics remains as a useful approximation at macroscopic scale and we will show in chapter 5 that quantum mechanics is consistent with classical mechanics.

There are two main types approaches to the quantum mechanics. One approach is approach the subject in chronological order of history: Look at the experiments that are inconsistent with classical mechanics in detail, then try to patch the existing theory following the footsteps of the pioneer physicists. Another approach is to look at the subject in hindsight, using our current existing knowledge and ask ourselves how to construct the theory in the most logical and concise manner. We shall take the latter route, as the history is full of wrong-footed steps and unnecessary confusion, our goal is to find the quickest route from complete ignorance of the theory to deep understanding of the principles of quantum mechanics.

Classically, if we want to describe the dynamics of the system we need a set of 6 numbers: position in 3 directions and momentum in 3 directions. If we know where the object is and how fast it is moving, we can predict its position and momentum in future by Newton's laws of motion.

At the scale of atoms and molecules, the system becomes extremely sensitive to the surrounding environments and becomes easily influenced by the random fluctuations which are normally negligible at the macroscopic scale. In addition, for us to gain knowledge of the system being studied, we interact with the system, which invariably disturbs the system. The disturbance may be again negligible at human scale but significant at small scale. Appreciating the complex nature of the physical state of an entity at small scale, we may need infinitely many numbers to fully encode the information of the object in the form of a probability distribution, due to the fundamental indeterminacy introduced by the disturbance of the system of interest. The concept of "**complete set of amplitudes**" $\{a_1, a_2, a_3, \cdots\}$ represents the full knowledge of the quantum entity we are interested in. The essence of the complete set of amplitude is that we can extract any information we want from the set, there is nothing left to know about the system apart from it.

When we have a list of numbers, it is natural to think of the list of numbers as a vector, where the component of the vector are the amplitudes in the complete set. The vector is effectively a container for all dynamical information of the system of interest. This vector has the dimension same as the number of amplitudes in the list, as the list may be infinitely long, the vector can be infinite dimensional. As these vectors are very different from the usual 3 dimensional vector we are familiar with, they have earned their own notation, we change the arrow above the letter to an angle bracket to the right $|\psi\rangle$. This is called a "ket vector" or simply a "**ket**". As this vector fully encodes the dynamical information of the system, it is also called a state vector. Thus, ket is the mathematical

representation of the physical state of the system, the **quantum state**. We have arrived at the first postulate of quantum mechanics:

The quantum state is represented by a ket.

These kets add and multiply by numbers like ordinary 3D vectors, thus they form their own vector space. We can also use our intuition for 3D vectors for these abstract state vectors. The length square of the vector is given by the sum of the modulus squares of all amplitudes as we allow amplitudes to be complex numbers.

$$\sum_i |a_i|^2 = 1 \tag{1.1}$$

The equation above is obtained by generalising the Pythagorean theorem to infinite dimensions. As we have encoded all the dynamical information into the components of the ket, the richness of the information is in the orientation of the ket in the infinite dimensional space we shall refer to as **Hilbert space** from now on. The length of the ket therefore encodes no information and can be set to one, in otherwords, we can scale the set up or down the entire set by a constant factor without loosing information, we will show how this is done later in this chapter. We also know the sum of probabilities of all possible outcomes of an event is one.

$$\sum_i P_i = 1 \tag{1.2}$$

As we have fixed the length of the ket to one, it will always remain one no matter what happens to the system. As the of probability will also remain one no matter what, we have established a connection between total probability and length of the state vector. We have used these amplitudes at the start to encode the information of the system in terms of a probability distribution, it is no surprise that there exist a close connection between amplitude and probability. Comparing the two equations above give us the modulus square of the amplitude is probability, this is called "**Born rule**".

$$\boxed{P_i = |a_i|^2} \tag{1.3}$$

We have arrived at the second postulate of quantum mechanics:

Probability is the modulus square of the amplitude.

Strictly speaking, postulates cannot be derived from existing facts, they should be accepted without question, all we should be concerned is whether it accounts for observations in the real world. However, I have deliberately tried to motivate these postulates to give you some intuitive satisfaction in the fundamental framework of quantum mechanics. A more rigorous approach is to state all the postulates at the start and construct the rest of the quantum mechanics from there. The Born rule is the essence of quantum mechanics, it introduces the concept of probability amplitude that is absent in any other branches of science. The rule has far reaching consequences and is the key to the success of quantum mechanics, without it, quantum mechanics would collapse.

It is natural for us to choose a basis for the ket representing the state of our system, and write the ket as a sum of the basis kets, This is analogous to writing a 3D vector as the linear combination of the unit vectors in x, y, z directions. A linear combination simply means multiplying each vector by a constant then adding the results. This sum is called "**quantum superposition**", another fundamental feature of quantum mechanics.

Quantum superposition is what ultimately give rise to the linear algebra framework of quantum mechanics.

$$|\psi\rangle = \sum_i a_i |i\rangle \tag{1.4}$$

To discover the physical meanings of the basis ket, we consider a concrete example: Allows a_i to be the amplitude of finding the particle in the ith potential well, where $|a_i|^2$ is the probability given by the Born rule. If we now set a_i to 1, the $|\psi\rangle$ becomes $|i\rangle$ and we are certain to find the particle in the ith well, hence we can conclude $|i\rangle$ is the state of of particle definitely being in the ith well. Our choice of potential well as an example is arbitrary, we equally could have used energy levels or any other measurable quantities to illustrate the same point. Thus the basis kets are states of certainty. It is reassuring that basis kets are the simplest possible states and they combine to give rise to a complicated state with uncertainty.

So far, we have encoded all the dynamical information in the ket, in which the amplitudes are the components of the vector and are closely related to a probability distribution that we seek after. So our next goal is to extract these amplitudes from the ket. In other words, we need a set of functions that input infinite dimensional vectors and output a complex numbers, which is the component in basis ket. We now introduce new notation for these functions of left angle bracket which is called a "bra vector" or simply a "**bra**", hence if we have left and right angle brackets strung together, we have a "bra-ket". This is known as the **Dirac notation**, and its power will be apparent shortly.

To proceed we ask the question of "What is the amplitude of the system being in the ith well given our system is in the jth well?" The answer is, being in different wells are mutually exclusive events, the amplitude is 1 if $i = j$ and 0 otherwise. We can define the **Kronecker delta** using this rule.

$$\delta_{ij} \equiv \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases} \tag{1.5}$$

We now have neat expression of the orthogonality and normalisation of definite states, collectively known as **orthonormality**.

$$\langle i|j\rangle \equiv \delta_{ij} \tag{1.6}$$

We use this equation to define the basis bra, any generic bra can be expressed as a linear combination the basis bras, just like any generic ket can be expressed as a linear combination basis kets. In fact, bras form its own vector space called the Hilbert **dual space**, which is a different space from the Hilbert space inhabited by kets. We confirm the number of basis bras must be the same as the number of basis kets by considering the dimensions of the two spaces.

A bra is a function thus it is only fully defined once every input ket has a output complex number. The simplest possible function is to map one basis ket onto itself but maps every other ket onto zero, which is the definition of the basis bra above. We can then construct any arbitrary function by scaling every single output of the basis bra then sum over all the basis. Thus we can conclude there are as many basis bras as basis kets. Thus for every ket, there is a corresponding bra.

The word "orthogonality" hints at the connection to orthogonality between vectors in 3D space, and there is indeed a close analogy between the two concepts. The standard

basis vectors in 3D space are mutually orthogonal to each other and normalised. Where $\hat{e}_i$ is the ith basis vector, conventionally $i = 1, 2, 3$ for x, y, z respectively.

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} \tag{1.7}$$

We now ask the question of "What is the amplitude of the system being in the ith well for an arbitrary state $|\psi\rangle$?", we show that it is mathematically represented by: $\langle i|\psi\rangle$ by expanding $|\psi\rangle$ in terms of its basis, which are definite states of being in a well, then use the linearity of bra to pass through the sum, as bra is a function on kets rather than numbers. The $\langle i|j\rangle$ is the Kronecker delta and vanishes unless $j = i$, therefore the sum contracts to a single number, a_i. Thus we have proved the amplitude of the system being in the ith definite state is given by $\langle i|\psi\rangle$, in other words if we wish to obtain the amplitude of the system being found in the ith definite state, we left multiply the system's ket by $\langle i|$. It is worth noting that we must use a different index for the sum, this is because the bra does not have to correspond any arbitrary basis ket that is being summed over, but may correspond to a specific basis ket.

$$\langle i|\psi\rangle = \langle i| \sum_j a_j |j\rangle = \sum_j a_j \langle i|j\rangle = \sum_j a_j \delta_{ij} \tag{1.8}$$

$$\boxed{a_i = \langle i|\psi\rangle} \tag{1.9}$$

Intuitively we know that the component of a arbitrary 3D vector in a direction vector is given by the projection onto the basis vector of that direction, mathematically given by the dot product.

$$a_i = \hat{e}_i \cdot \vec{r} \tag{1.10}$$

Compare equations (1.6) with (1.7) and (1.9) with (1.10) shows that the bra-ket has a close connection with the dot product in 3D space. We therefore generlise the dot product to space that may be infinite dimension to the "**inner product**" and the angle brackets are in fact borrowed from the notation of inner product in maths, vertical bar can be interpreted as "given" in conditional probability.

The dot product of a vector with itself gives the square of the length of the vector, by analogy, the inner product of a ket with itself is also the length square of the ket, which must be greater than or equal to zero. Thus given an expression for ket as a sum of its basis, we must define the amplitudes in the bra to be complex conjugate of the kets, otherwise lengths may not be positive or even real.

$$\langle \psi| = \sum_i a_i^* \langle i| \tag{1.11}$$

If we have two different kets defined as below:

$$|\psi\rangle = \sum_i a_i |i\rangle \tag{1.12}$$

$$|\phi\rangle = \sum_i b_i |i\rangle \tag{1.13}$$

Then the two possible inner products are:

$$\langle \phi | \psi \rangle = \sum_i b_i^* a_i \tag{1.14}$$

$$\langle \psi | \phi \rangle = \sum_i a_i^* b_i \tag{1.15}$$

Taking the complex conjugate of RHS (1.14) gives RHS of (1.15), this means the complex conjugate of an inner product is the inner product with left and right argument swapped.

$$\boxed{\langle \phi | \psi \rangle^* = \langle \psi | \phi \rangle} \tag{1.16}$$

At the start we said the ket representing our physical system must have unit length. If a ket does not have length of 1 to start with, we can always noramlise it by dividing the ket with the square root of the inner product with itself, which is analogous to how we would noramlise a 3D vector.

$$|\psi'\rangle = \frac{|\psi\rangle}{\sqrt{\langle \psi | \psi \rangle}} \tag{1.17}$$

$$\hat{n} = \frac{\vec{r}}{\sqrt{\vec{r} \cdot \vec{r}}} \tag{1.18}$$

Everything we have discussed so far assumes a descrete spectrum, such as individual potential wells and energy levels. However, some quantities are not descrete, for example, position of a free particle is a contineous quantity. To incoporate this into our mathematical framework, we note that integral is the continuous analogue of a sum. We write a generic state $|\psi\rangle$ as a superposition of definite states of position $|x\rangle$ as the following integral:

$$|\psi\rangle = \int dx\, \psi(x) |x\rangle \tag{1.19}$$

Where the coefficient is now a function:

$$\boxed{\psi(x) = \langle x | \psi \rangle} \tag{1.20}$$

The coefficient function is called the **wavefunction**. The modulus square of the wavefunction is now the probability density instead of probability in the discrete analogue.

$$P(x) = |\psi(x)|^2 \tag{1.21}$$

The inner product is by analogy the integral:

$$\langle \phi | \psi \rangle = \int dx\, \phi^*(x) \psi(x) \tag{1.22}$$

The Kronecker delta expressing the orthonoramlity of basis states is now replaced by **Dirac delta**.

$$\langle x | x' \rangle = \delta(x - x') \tag{1.23}$$

$$\delta(x - x') = \begin{cases} +\infty, & \text{if } x = x', \\ 0, & \text{if } x \neq x'. \end{cases} \tag{1.24}$$

Dirac delta is a rather strange function, we will look at this function in more detail in the next chapter.

A vector can be described by various coordinate systems, two of the most common are Cartesian and polar coordinate systems. Each coordinate system assigns a different set of numbers to describe the same vector, thus the description of the vector itself is not unique. A ket can also be described by various ways by choosing different basis kets. **Representation** is the choice of basis kets, the quantum state itself is independent of representation. Position representation is when we choose definite state of position $|x\rangle$ as the basis kets, momentum representation is when we choose definite state of momentum $|p\rangle$ as the basis kets, so on so forth.

$$|\psi\rangle = \int dp\, \tilde{\psi}(p)|p\rangle \tag{1.25}$$

A physical system has three ingredients: States, observables and dynamics. We have now understood how quantum mechanics describes the state of a system. In the next part, we will begin to discuss how the observables are represented in quantum mechanics.

2 Operator

In chapter 1, we introduced ket, which is a special kind of vector that may be infinite dimensional and inhabits the Hilbert space. In linear algebra, another important object is the matrix. Matrix act on a vector to give another vector. We can abstract the concept of matrix, so they can now act on kets rather than ordinary vectors. Turns out, matrix can be regarded as a representation of **linear operator** in a given basis. We choose the unit vectors in x, y, z directions as standard basis in 3D, hence matrices operating in 3D space are always assumed to be in standard basis. There is no standard basis in quantum mechanics, hence the mathematical framework of quantum mechanics is abstract linear algebra. In chapter 1, we have associated kets with the quantum states, we shall now discover the physical significance of linear operator in quantum mechanics.

An operator $\hat{Q}$ is linear if it satisfies the following equation:

$$\hat{Q}(a|\psi\rangle + b|\phi\rangle) = a\hat{Q}|\psi\rangle + b\hat{Q}|\phi\rangle \tag{2.1}$$

Now consider the following expansion of an arbitrary quantum state:

$$|\psi\rangle = \sum_i a_i|i\rangle = \sum_i |i\rangle a_i = \sum_i |i\rangle\langle i|\psi\rangle \tag{2.2}$$

$$|\psi\rangle = \left(\sum_i |i\rangle\langle i|\right)|\psi\rangle \tag{2.3}$$

The sum inside the round bracket in the last line operating on the state returns the same state, it is analogous to multiplying any vector by the identity matrix. Thus we have found the expression for the **identity operator** in quantum mechanics:

$$\boxed{\hat{I} = \sum_i |i\rangle\langle i|} \tag{2.4}$$

Following the same argument but for continuous spectrum yields:

$$\hat{I} = \int dx\, |x\rangle\langle x| \tag{2.5}$$

Notice that we now have a "ket-bra", this is known as the "**outer product**", this has the two parts of the angular bracket in reverse order to the "inner product". All outer products are linear operators whereas all inner products are numbers.

We are now in a position to gain more insight into the nature of Dirac delta, as it is not a function in the traditional sense. We begin by stating the wavefunction for a specific position x' in Bra-Ket notation, then insert the identity operator.

$$\langle x'|\psi\rangle = \psi(x') \tag{2.6}$$

$$\langle x'|\Big(\int dx\, |x\rangle\langle x|\Big)|\psi\rangle = \psi(x') \tag{2.7}$$

$$\int dx\, \langle x'|x\rangle\langle x|\psi\rangle = \psi(x') \tag{2.8}$$

$$\int dx\, \delta(x' - x)\psi(x) \equiv \psi(x') \tag{2.9}$$

We discover that just like the Kroncker delta, Dirac delta has the same sieving property that picks out the parameter that is not being summed or integrated over. We use this property to define the Dirac delta. We gain further insight by noticing that the integrand vanishes unless $x = x'$ so we can take $\psi(x')$ out of the integral.

$$\psi(x') \int dx\, \delta(x' - x) = \psi(x') \tag{2.10}$$

$$\int dx\, \delta(x' - x) = 1 \tag{2.11}$$

Thus Dirac delta integrates to one, just like a probability distribution. This is remarkable, as despite that the function vanishes at every point except $x = x'$, there is still unit area under the curve. Thus we have demonstrated the normalising property of Dirac delta. It is useful to think of Dirac delta as a function with the limit of being a infinitely narrow and tall, for example, a Gaussian function with the width tending toward zero. Statistically, this is a normal distribution with standard deviation tending towards zero, representing the increasing certainty of measurement being very close to the mean. The fact that the function is not well-behaved reflects the fact the these definites states of a continuous parameter $|x\rangle$ are idealised mathematical constructs and are physically unobtainable. However, despite that these states can never be obtained, they remain a crucial object in the mathematical framework of quantum mechanics.

To determine the physical quantities of a system such as position or energy, we need to perform measurements. Any physical quantities that can be measured is called "**Observable**".

Suppose we have an linear operator that "identifies" the definite states of that observable and then attaches the value of the physical quantity to it, then this linear operator is the repository of all the information regarding that observable, we have arrived at our third postulate:

Observable is represented by linear operator.

The justification for this is that all information of the state of the system is encoded in the "orientation" of the ket in the Hilbert space, and only certain kets are definite states of a given observable. In return, these special kets that only get stretched after the linear operator has been applied and the stretch factor defines the linear operator.

Consider the following equation:

$$\hat{E}|E\rangle = E|E\rangle \tag{2.12}$$

Where $\hat{E}$ is the linear operator representing the energy observable, $|E\rangle$ is a definite state of energy conveniently labelled by - E the energy of the definite state, which is a real number. This is a rather confusing notation, thus we change to a clearer notation, with discrete spectrum:

$$\boxed{\hat{H}|n\rangle = E_n|n\rangle} \tag{2.13}$$

This is the **Schrödinger equation** (Time-independent), we shall refer to this equation as **TISE**. This type of equation where the effect of the linear operator merely scales the special states of interest is called eigenvalue equation and most of quantum mechanics

is centered around solving this equation, analogous to how classical mechanics centers around solving Newton's second law. The solutions of TISE are the eigenvalues and eigenkets, which physically corresponds to energy spectrum and definite states of energy, we will see in chapter 5 how these solutions are connected to the time evolution of a quantum state. $|n\rangle$ are the **eigenstates**, which are now labelled by **quantum number** n, which is the energy level, as we are interested in solving systems with quantised energy most of the time. E_n is the energy of the nth energy level, the **eigenvalue**. We have also relabelled our energy operator to $\hat{H}$ which is named after W. R. Hamilton, who has historical importance to the development of quantum mechanics. We shall also rename our energy operator to **Hamiltonian** to honor him.

We now think about what happens after we have made a measurement on a system. Suppose we want to measure the energy of a particle, if our measurement has arbitrary precision then we become certain about the energy of the particle, thus we can say the state is $|n\rangle$ with energy E_n. We are certain in the sense that a subsequent measurement on energy will definitely also be $|n\rangle$. We have arrived at the fourth postulate of quantum mechanics:

The result of measurement gives one of the eigenvalues and the state is updated to the eigenstate of that observable.

It is tempting to think that the update of the state is merely a reflection of us has gained knowledge of the system, however, it is important to note that the ket represents the physical state of the system rather than just our subjective understanding of it. Update of the ket reflects a physical disturbance of the system, a measurement simultaneously gains us knowledge and disturbs the system, this is the essence of quantum mechanics.

However, quantum mechanics is silent regarding how the state is updated from a generic state $|\psi\rangle$ to an eigenstate, this phenomenon is known as the **collapse of the wavefunction**, and is one of the unsolved questions in physics. For now, we should not be concerned with the underlying mechanism of the collapse, as we simply regard this as part of the postulate.

We deduce the expanded form of the operator by using it on a generic state and then expand the state as a sum of the eigenstates of the operator:

$$\hat{H}|\psi\rangle = \hat{H}\sum_n a_n|n\rangle = \sum_n a_n\hat{H}|n\rangle = \sum_n a_n E_n|n\rangle = \sum_n \langle n|\psi\rangle E_n|n\rangle \tag{2.14}$$

$$\hat{H}|\psi\rangle = \left(\sum_n E_n|n\rangle\langle n|\right)|\psi\rangle \tag{2.15}$$

Thus we conclude:

$$\hat{H} = \sum_n E_n|n\rangle\langle n| \tag{2.16}$$

We have only considered energy as an observable but the procedure above can be generalised to any observable. Suppose $\hat{R}$ is one of the observables, it would have the eigenvalue equation and expansion:

$$\boxed{\hat{R}|r_i\rangle = r_i|r_i\rangle} \tag{2.17}$$

$$\boxed{\hat{R} = \sum_i r_i |r_i\rangle\langle r_i|} \tag{2.18}$$

Generalising to continuous spectrum yields:

$$\hat{x} = \int dx\, x|x\rangle\langle x| \tag{2.19}$$

We now take some time to develop the notation further. As the operator is linear, another ket is always generated after the application of the operator to the ket. We can name this new ket anything we like, but it is often useful to label the ket in a way such that we can tell what operator has been used to generate this new ket:

$$\hat{Q}|\phi\rangle = |\hat{Q}\phi\rangle \tag{2.20}$$

It will be proved to be useful to have an operator that can operate on a dual bra of the original ket to produce the dual bra of the resultant ket on the right side of equation (2.20). We use a dagger † to denote the relationship between this new operator to the original operator $\hat{Q}$.

$$\langle\phi|\hat{Q}^\dagger = \langle\hat{Q}\phi| \tag{2.21}$$

We now right multiply both sides by $|\psi\rangle$ to define the **Hermitian adjoint** of an operator:

$$\langle\phi|\hat{Q}^\dagger\psi\rangle \equiv \langle\hat{Q}\phi|\psi\rangle \tag{2.22}$$

Taking the complex conjugate of both sides yields:

$$\langle\hat{Q}^\dagger\psi|\phi\rangle = \langle\psi|\hat{Q}\phi\rangle \tag{2.23}$$

The two equations above gives us a simple rule for manipulating Hermitian adjoints: For an operator to enter the left argument of the inner product from the right, it must loose or gain a dagger. Same rule also applies for leaving the left argument going to the right.

Taking the double complex conjugate of the RHS of definition (2.22) yields a useful equation:

$$\langle\phi|\hat{Q}^\dagger|\psi\rangle = \langle\psi|\hat{Q}|\phi\rangle^* \tag{2.24}$$

Choosing the ket and the bra from an orthonormal basis by setting $|\psi\rangle = |j\rangle$ and $\langle\phi| = \langle i|$ yields:

$$Q^\dagger_{ij} = Q^*_{ji} \tag{2.25}$$

Where matrix element is defined as:

$$Q_{ij} \equiv \langle i|\hat{Q}|j\rangle \tag{2.26}$$

This demonstrates that the Hermitian adjoint in matrix representation is just the conjugate transpose.

Physical intuition suggests observable must have 3 important properties:

1. It has real eigenvalues.

2. It has orthogonal eigenstates.

3. Its eigenstates form a complete set.

The justifications are as follows: For property 1, physical quantities must have real values. For property 2, states of certainty of different values of a physical quantity are mutually exclusive events. Thus the kets representing each definite outcome must point in completely different directions.

A **Hermitian operator** satisfies the following equation:

$$\hat{R} = \hat{R}^\dagger \tag{2.27}$$

Turns out, a Hermitian operator can satisfy all three properties simultaneously. We show the first two properties are true by left multiplication of the eigenvalue of equation (2.17) by $\langle r_j|$ to yield:

$$\langle r_j|\hat{R}|r_i\rangle = r_i\langle r_j|r_i\rangle \tag{2.28}$$

We then swap the index i and j and take the complex conjugate of both sides:

$$\langle r_i|\hat{R}|r_j\rangle^* = r_j^*\langle r_i|r_j\rangle^* \tag{2.29}$$

Using the the fact that $\hat{R}$ is Hermitian we can equate the LHS of the two equations above, the RHS also equates:

$$r_i\langle r_j|r_i\rangle = r_j^*\langle r_i|r_j\rangle^* \tag{2.30}$$

Complex conjugate of an inner product is the inner product with left and right entry swapped, then slight rearrangement yields:

$$0 = (r_j^* - r_i)\langle r_j|r_i\rangle \tag{2.31}$$

Consider if $i = j$, then the inner product is the square length, as we assumed the state must exist so it must have a non-zero length. The difference between two eigenvalues must vanish and we can conclude:

$$r_i = r_i^* \tag{2.32}$$

Only real numbers are their own complex conjugate, therefore the eigenvalues of a Hermitian operator are real. This completes the proof of the first property of the observable.

Consider if $i \neq j$, then the difference does not vanish and we can conclude:

$$\langle r_j|r_i\rangle = 0 \tag{2.33}$$

Only orthogonal states have zero inner product provided that non of the kets vanish. This completes the proof of the second property of the observable.

The proof for the third property is beyond the scope of this chapter, but our physical intuition dictates it must be satisfied by the Hermitian operator. The eigenkets of the observable must form a complete basis so that any generic state of the system can be expressed as a sum of all basis state. If the eigenkets does not form a complete basis, projections onto certain directions would be undefined and the measurement of a physical

quantity such as energy has the probability of not returning a result. As it is impossible for the measuring instrument to ignore our request when it is not malfunctioning, we conclude the operator representing observable must have eigenkets that forms a complete basis set in the sense that it spans the entire Hilbert space.

Thus observable is represented by Hermitian operator in quantum mechanics.

We now establish some rules for Hermitian adjoint which will be useful in future:

1.
$$\hat{Q}^{\dagger\dagger} = \hat{Q} \tag{2.34}$$

2.
$$(\hat{Q} + \hat{R})^\dagger = \hat{Q}^\dagger + \hat{R}^\dagger \tag{2.35}$$

3.
$$(\hat{Q}\hat{R})^\dagger = \hat{R}^\dagger \hat{Q}^\dagger \tag{2.36}$$

4.
$$|\psi\rangle^\dagger = \langle\psi| \tag{2.37}$$

5.
$$c^\dagger = c^* \tag{2.38}$$

Rule 1 and 2 can be proved easily by applying the definition of Hermitian adjoint and are left as an exercise for the reader. Rule 4 can be treated as the definition of bra and also implies: $\langle\psi|^\dagger = |\psi\rangle$, by applying rule 1. Furthermore, in matrix representation, ket is a column vector and bra is a row vector, normal rules for matrix multiplication shows the bra-ket is a number and ket-bra is a matrix. To prove rule 5, we note that the matrix representation of a number is the number itself, and the transpose has no effect. We are just left with the complex conjugate. Note that the first three rules also applies when any linear operator is replaced by kets, bras or numbers.

We shall now prove rule 3:

$$\langle\phi|(\hat{Q}\hat{R})^\dagger|\psi\rangle = \langle\psi|\hat{Q}\hat{R}|\phi\rangle^* = \langle\hat{Q}^\dagger\psi|\hat{R}\phi\rangle^* = \langle\hat{R}\phi|\hat{Q}^\dagger\psi\rangle = \langle\phi|\hat{R}^\dagger\hat{Q}^\dagger|\psi\rangle \tag{2.39}$$

As both $|\psi\rangle$ and $\langle\phi|$ are arbitrary, this completes the proof.

It is useful to define functions of operators as:

$$f(\hat{R}) \equiv \sum_i f(r_i)|r_i\rangle\langle r_i| \tag{2.40}$$

We will see later than some operators are functions of other operators, for example, the kinetic and potential energy operators.

At the end of this chapter, we make our first connection back to classical mechanics by realising that the physical quantities we measure at the macroscopic scale is given by the **expectation value** of the observable given by:

$$\boxed{\langle\hat{Q}\rangle = \langle\psi|\hat{Q}|\psi\rangle} \tag{2.41}$$

Where $\hat{Q}$ is an arbitrary observable, represented by a Hermitian operator.

The proof runs as follows:

$$\langle\psi|\hat{Q}|\psi\rangle = \langle\psi|\left(\sum_i q_i|q_i\rangle\langle q_i|\right)|\psi\rangle = \sum_i q_i\langle\psi|q_i\rangle\langle q_i|\psi\rangle \tag{2.42}$$

$$= \sum_i q_i|a_i|^2 = \sum_i q_i P_i \equiv \langle\hat{Q}\rangle \tag{2.43}$$

Where we have used the definition of expectation value in the last equality. Classical mechanics can therefore be regarded as the mechanics of expectation values, it is only concerned with the mean that the true physical quantities fluctuate about.

3 Commutator

Commutator is a crucial object in the mathematical framework of quantum mechanics. As we have decided to use linear operators to represent observable, the inability for two operators to commute has far reaching consequences on the result of measurement. In this chapter, we will see how the commutator ultimately leads to the uncertainty principle. In addition, it is an important tool for quantum mechanical calculations.

Commutator is defined as:

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A} \tag{3.1}$$

When the commutator between two operator is zero, it is said that the two operators **commute**. The following rules will be useful when manipulating commutators:

1.
$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}] \tag{3.2}$$

2.
$$[\hat{A} + \hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}] \tag{3.3}$$

3.
$$[\hat{A}\hat{B}, \hat{C}] = [\hat{A}, \hat{C}]\hat{B} + \hat{A}[\hat{B}, \hat{C}] \tag{3.4}$$

4.
$$[\hat{A}, f(\hat{B})] = [\hat{A}, \hat{B}]\frac{d\hat{f}}{dB} \tag{3.5}$$

Rule 1 to 3 can be proved by substituting in definition of commutator and are left as an exercise for the reader. It is worth noting rule 3 is reminiscent of the product rule for differentiation. It is also worth mentioning that all operators commute with itself and functions of itself and with numbers. In addition, operators operating in different degrees of freedom always commute.

We shall now prove rule 4:

We assume the function f has a convergent Taylor series so that: $f = f_0 + f'\hat{B} + \frac{1}{2}\hat{B}^2\cdots$, where $f_0 \equiv f(0)$, $f' \equiv (df/dx)_0\cdots$, then expand out the commutator using rule 2 and 3 yields:

$$\begin{aligned}[\hat{A}, f(\hat{B})] =& f'(0)[\hat{A}, \hat{B}] + \frac{1}{2}f''(0)([\hat{A}, \hat{B}]\hat{B} + \hat{B}[\hat{A}, \hat{B}]) \\ &+ \frac{1}{3!}f'''(0)([\hat{A}, \hat{B}]\hat{B}^2 + \hat{B}[\hat{A}, \hat{B}]\hat{B} + \hat{B}^2[\hat{A}, \hat{B}]) + \cdots\end{aligned} \tag{3.6}$$

In the case in which $\hat{B}$ commutes with $[\hat{A}, \hat{B}]$, the equation simplifies to:

$$[\hat{A}, f(\hat{B})] = [\hat{A}, \hat{B}](f' + f''\hat{B} + \frac{1}{2}f'''\hat{B}^2 + \cdots) \tag{3.7}$$

Recognising the last line is the Taylor series expansion of df/dB yields (3.5), this completes the proof.

The most important consequence when two obsevables commute is that there exist a complete set of mutual eigenstates, such that any state of the system can be expressed as a

superposition of these mutual eigenstates. Conversely, if two operators doesn't commute, there doesn't exist a complete set of mutual eigenstates.

Suppose we have a complete set of eigenstates of $\hat{A}$ such that it satisfies the following equation:

$$\hat{A}|a\rangle = a|a\rangle \tag{3.8}$$

We now left multiply both sides of the equation by $\hat{B}$, which commutes with $\hat{A}$:

$$\hat{B}\hat{A}|a\rangle = a\hat{B}|a\rangle \tag{3.9}$$

$$\hat{A}|\hat{B}a\rangle = a|\hat{B}a\rangle \tag{3.10}$$

Therefore $\hat{B}|a\rangle$ is also an eigenket of $\hat{A}$ with eigenvalue a. If the eigenstates are non-degenerate for the same eigenvalue, then we can conclude:

$$\hat{B}|a\rangle = b|a\rangle \tag{3.11}$$

Where b is the proportionality constant also the eigenvalue. It is often useful to rebel the simultaneous eigenstates with both eigenvalues. $|a\rangle \to |a, b\rangle$

We have shown that commuting observable has a complete set of mutual eigenstates for the non-degenerate eigenvalue. The result can also be generalised to degenerate case. This is a powerful result and has far reaching consequences on measurement.

We shall now derive the uncertainty principle.

The expectation value of the commutator of two arbitrary Hermitian operators $\hat{Q}$ and $\hat{R}$ is given by:

$$\langle\psi|[\hat{Q}, \hat{R}]|\psi\rangle = \langle\psi|\hat{Q}\hat{R}|\psi\rangle - \langle\psi|\hat{R}\hat{Q}|\psi\rangle = \langle\hat{Q}\psi|\hat{R}\psi\rangle - \langle\hat{R}\psi|\hat{Q}\psi\rangle \tag{3.12}$$

$$= \langle\hat{Q}\psi|\hat{R}\psi\rangle - \langle\hat{Q}\psi|\hat{R}\psi\rangle^* = 2i\Im\langle\hat{Q}\psi|\hat{R}\psi\rangle \tag{3.13}$$

We now use the following facts from elementary mathematics: For any complex number c and vectors $\vec{a}$ and $\vec{b}$:

$$|c| \geq |\Im(c)| \tag{3.14}$$

$$|\vec{a}||\vec{b}| \geq |\vec{a} \cdot \vec{b}| \tag{3.15}$$

Generalising the inequality for vectors to kets and combining the two inequalities above and use (3.15) to replace $\Im\langle\hat{Q}\psi|\hat{R}\psi\rangle$ yields:

$$\sqrt{\langle\hat{Q}\psi|\hat{Q}\psi\rangle}\sqrt{\langle\hat{R}\psi|\hat{R}\psi\rangle} \geq |\langle\hat{Q}\psi|\hat{R}\psi\rangle| \geq \left|\frac{1}{2i}\langle\psi|[\hat{Q}, \hat{R}]|\psi\rangle\right| \tag{3.16}$$

$$\sqrt{\langle\hat{Q}^2\rangle}\sqrt{\langle\hat{R}^2\rangle} \geq \frac{1}{2}|\langle[\hat{Q}, \hat{R}]\rangle| \tag{3.17}$$

We now make the following replacements:

$$\hat{Q} = \delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle \tag{3.18}$$

$$\hat{R} = \delta\hat{B} \equiv \hat{B} - \langle\hat{B}\rangle \tag{3.19}$$

We also note that the commutator is invariant under additive constants, as constants commute with everything:

$$\sqrt{\langle\delta\hat{A}^2\rangle}\sqrt{\langle\delta\hat{B}^2\rangle} \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle| \tag{3.20}$$

We define uncertainty as the root-mean-square deviation:

$$\sigma_A \equiv \sqrt{\langle\delta\hat{A}^2\rangle} \tag{3.21}$$

The result is the generalised **uncertainty principle**:

$$\boxed{\sigma_A\sigma_B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|} \tag{3.22}$$

When the operators representing the two observables do not commute, it is therefore impossible to simultaneously and precisely measure the two physical quantities, no matter how accurate the measuring instrument is. When the two operators commute, it is theoretically possible to determine the two physical quantities simultaneously and precisely.

One of the most important equation in quantum mechanics is the canonical commutation relation, which we shall derive in the next chapter:

$$[\hat{x},\hat{p}] = i\hbar \tag{3.23}$$

Substitution into the generalised uncertainty principle directly leads to the **Heisenberg uncertainty principle**:

$$\boxed{\sigma_x\sigma_p \geq \frac{\hbar}{2}} \tag{3.24}$$

To see the uncertainty principle in physical space, we need to work with wavefunctions. We now take another result we will prove in the next chapter, the momentum operator in position representation:

$$\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{\partial\langle x|\psi\rangle}{\partial x} \tag{3.25}$$

It is left as an exercise for the reader to check the equation (3.27) is consistent with the canonical commutation relation (3.25).

An interesting case is when the state is a definite state of momentum $|p\rangle$, the equation then becomes a first order differential equation:

$$p\langle x|p\rangle = -i\hbar\frac{\partial\langle x|p\rangle}{\partial x} \tag{3.26}$$

Where we have used the fact that $|p\rangle$ is an eigenket of $\hat{p}$. We see $\langle x|p\rangle$ as a function of the left argument of inner product. The differential equation is solved by separation of variables:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar} \tag{3.27}$$

Taking the complex conjugate of both sides yields another useful equation:

$$\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \tag{3.28}$$

The wavefunction cannot be normalised in the traditional sense, the normalisation factor turns out to be $1/\sqrt{2\pi\hbar}$ and the analysis is beyond the scope of the chapter, but also currently of no interest to us. Recalling probability density function is given by the modulus square of the wavefunction, immediately, we can tell a particle with definite amount of momentum is equally likely to be anywhere in space as $|\langle x|p\rangle|^2$ is independent of x. Similarly, a particle with a definite position is equally likely to have any momentum as $|\langle p|x\rangle|^2$ is independent of p, this is obviously physically unobtainable just like how we deduced this from the property of Dirac delta back in chapter 2.

It is physically impossible to have measuring instrument of infinitely high precision, so let us now consider a more realistic case. We want to consider a more general relationship between $\langle p|\psi\rangle$ and $\langle x|\psi\rangle$, where $|\psi\rangle$ can be any arbitrary state. We begin by inserting the identity operator in position basis, then use equation (3.30).

$$\langle p|\psi\rangle = \int dx\, \langle p|x\rangle\langle x|\psi\rangle \tag{3.29}$$

$$\langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int dx\, \langle x|\psi\rangle e^{-ipx/\hbar} \tag{3.30}$$

Hence, the momentum basis wavefunction is just the **Fourier transform** of position basis wavefunction. Same argument also yields:

$$\langle x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int dp\, \langle p|\psi\rangle e^{ipx/\hbar} \tag{3.31}$$

Thus position basis wavefunction is the inverse transform of momentum basis wavefunction. Let's now look at a concrete example: For a particle with a normally distributed position about the origin, with wavefunction:

$$\langle x|\psi\rangle = \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x^2/4\sigma^2} \tag{3.32}$$

Where we recognise standard deviation σ as the uncertainty in position σ_x.

After some very tedious algebra, which is left as an exercise for the reader, we obtain the momentum basis wavefunction:

$$\langle p|\psi\rangle = \frac{1}{(2\pi\hbar^2/4\sigma^2)^{1/4}} e^{-p^2\sigma^2/\hbar^2} \tag{3.33}$$

Hence, the momentum of the particle is also normally distributed, with $\sigma_p = \hbar/2\sigma$, this further suggests the product of uncertainties in position and momentum is exactly $\hbar/2$. This is the minimum threshold of Heisenberg uncertainty principle (3.26), and the product of uncertainty always exceeds this minimum value for other types of distributions. We can now understand the trade-off between the uncertainties in position and momentum from the perspective of: To localise a particle in space, we need wavefunctions of different momentum thus increasing the uncertainty in momentum, vice versa if we want the particle to have a small uncertainty in momentum the particle's position becomes spread out in space.

4 Generator

In chapter 2, we introduced operator as the repository of information regarding the observable it represents. We now discover what the operator does to the quantum state when it operates.

We have been using $|\psi\rangle$ to represent the quantum state, it must have unit length as the total probability of measurement of any physical quantity must always be one.

$$\langle\psi|\psi\rangle = 1 \tag{4.1}$$

We now allow the state to evolve to some other state $|\psi'\rangle$, by mathematically applying some linear operator $\hat{U}$. Length of the new ket must remain to be one.

$$|\psi'\rangle = \hat{U}|\psi\rangle \tag{4.2}$$

$$\langle\psi'|\psi'\rangle = \langle\psi|\hat{U}^\dagger\hat{U}|\psi\rangle = 1 \tag{4.3}$$

Hence we can conclude:

$$\hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \hat{I} \tag{4.4}$$

An operator $\hat{U}$ is said to be an **unitary operator** if it satisfies the two equations above. An useful geometric view is that the unitary operator rotates the ket in Hilbert space.

We now parametrise $\hat{U}$ with real parameter θ, the parameter quantifies how much the state has changed. Taylor series expansion of the operator yields:

$$\hat{U}(\theta) = \hat{U}(0) + \hat{U}'(0)\theta + \mathcal{O}(\theta^2) \tag{4.5}$$

Where $\hat{U}'(0)$ is the first derivative of $\hat{U}$ with respect to θ evaluate at $\theta = 0$. $\mathcal{O}(\theta^2)$ denotes the terms that are second order and higher. We note that $\hat{U}$ changes the state by an amount according to the value of the parameter θ, when $\theta = 0$, the state must remain unchanged, and we can conclude:

$$\hat{U}(0) = \hat{I} \tag{4.6}$$

Substitute into the previous equation yields:

$$\hat{U}(\theta) = \hat{I} + \hat{U}'\theta + \mathcal{O}(\theta^2) \tag{4.7}$$

Taking the Hermitian adjoint of both sides of (4.7), note that the identity operator is Hermitian and the parameter is real:

$$\hat{U}^\dagger(\theta) = \hat{I} + \hat{U}'^\dagger\theta + \mathcal{O}(\theta^2) \tag{4.8}$$

We now multiply the expansions of the unitary operator (4.8) and its Hermitian adjoint (4.8) and equating the result to identity, as $\hat{U}$ is unitary. We also make θ sufficiently small to $\delta\theta$ such that and second and higher order terms are negligible, then comparing the terms that is first order in $\delta\theta$ yields:

$$(\hat{U}'^\dagger + \hat{U}')\delta\theta = 0 \tag{4.9}$$

Provided that $\delta\theta \neq 0$ we conclude:

$$\hat{U}' = -\hat{U}'^{\dagger} \tag{4.10}$$

If we set $\hat{U}' = i\hat{\tau}$ then $\hat{\tau}$ is always Hermitian:

$$i\hat{\tau} = -(i\hat{\tau})^{\dagger} = -i^{*}\hat{\tau}^{\dagger} = i\hat{\tau}^{\dagger} \tag{4.11}$$

$$\hat{\tau} = \hat{\tau}^{\dagger} \tag{4.12}$$

Hence, for very small changes changes of the state, we can write the unitary operator as below, the minus sign is by convention.

$$\hat{U}(\delta\theta) = \hat{I} - i\delta\theta\hat{\tau} \tag{4.13}$$

An appreciably large change is achieved by large amount of small changes:

$$\hat{U}(\theta) = \lim_{N\to\infty} \hat{U}(\delta\theta)^{N} = \lim_{N\to\infty} \left(\hat{I} - i\frac{\theta}{N}\hat{\tau}\right)^{N} \equiv e^{-i\theta\hat{\tau}} \tag{4.14}$$

Where we have used $\delta\theta = \theta/N$ and the definition of exponential function. We have established an important result:

$$\boxed{\hat{U}(\theta) = e^{-i\theta\hat{\tau}}} \tag{4.15}$$

Right multiply both sides by $|\psi\rangle$ then differentiating with respect to θ and rearrange yields:

$$\boxed{i\frac{\partial|\psi\rangle}{\partial\theta} = \hat{\tau}|\psi\rangle} \tag{4.16}$$

Since $\hat{\tau}$ is Hermitian, so it may represent an observable, and this is indeed true. We have discovered that the operator representing the observable is not only the repository of the spectrum and definite states of that physical quantity but they have another role as **generator** of $\hat{U}$ and the transformation that $\hat{U}$ achieves. The Hermitian operator $\hat{\tau}$ gives the rate of change with respect to the parameter θ.

Turns out, there is a deep connection between conservation laws and continuous symmetry as stated in **Noether's theorem**:

For every continuous symmetry, we can associate a conservation law.

Newton's first law states that provided there is no external force, momentum is conserved under translation. Thus we associate momentum conservation with translational symmetry and define momentum as the generator of translation, we will justify this choice further in chapter 5. The translation operator can therefore be written as:

$$\boxed{\hat{T}(a) = e^{-ia\hat{p}/\hbar}} \tag{4.17}$$

Where a is distance displaced and $\hbar$ in the denominator ensures the exponent is dimensionless as exponentiation of unit is undefined. It is physically plausible that the translation operator is an exponential operator in displacement, as for sequential translations, operators multiply but displacements add, which is exactly the property of an exponential function.

$$\boxed{i\hbar\frac{\partial|\psi\rangle}{\partial a} = \hat{p}|\psi\rangle} \tag{4.18}$$

When the momentum operator acts on a state, it gives the the rate of change at which the state changes as we translate the system in space.

We now begin to derive a very fundamental result in quantum mechanics: the canonical commutation relation.

We begin by making the following observation: After the system has been displaced by some small distance δa:

$$\langle\hat{x}\rangle' = \langle\hat{x}\rangle + \delta a \tag{4.19}$$

$$\langle\psi'|\hat{x}|\psi'\rangle = \langle\psi|\hat{x}|\psi\rangle + \langle\psi|\delta a|\psi\rangle \tag{4.20}$$

$$\langle\psi|\hat{T}^\dagger(\delta a)\hat{x}\hat{T}(\delta a)|\psi\rangle = \langle\psi|\hat{x} + \delta a|\psi\rangle \tag{4.21}$$

$$\Big(\hat{I} + \frac{i\delta a\hat{p}}{\hbar}\Big)\hat{x}\Big(\hat{I} - \frac{i\delta a\hat{p}}{\hbar}\Big) = \hat{x} + \delta a \tag{4.22}$$

Equating the term that is first order in δa yields:

$$\frac{i}{\hbar}\delta a\hat{p}\hat{x} - \frac{i}{\hbar}\delta a\hat{x}\hat{p} = \delta a \tag{4.23}$$

Cancel δa on both sides and using the definition of commutator yields:

$$\frac{i}{\hbar}[\hat{p}, \hat{x}] = 1 \tag{4.24}$$

Multiply both sides by $i\hbar$ and use the negative sign to swap the order of arguments in the commutator yields the **canonical commutation relation**:

$$\boxed{[\hat{x}, \hat{p}] = i\hbar} \tag{4.25}$$

We have seen at the end of the last chapter that the Heisenberg uncertainty principle is a direct consequence of this commutator. We can use the fact that operators operating in different degrees of freedom commute to generalise this result to 3 dimensions. It is left as an exercise for the reader to derive this commutation relation in 3 dimensions by considering translation in 3D space.

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij} \tag{4.26}$$

It is often easier to work with wavefunctions in physical space rather than abstract state vectors in Hilbert space. The most useful representation is often the position representation, we shall now derive the form of the momentum operator in position representation as we have promised in chapter 3.

We begin by Taylor series expanding the wavefunction around x:

$$\psi(y) = \sum_{k=0}^{\infty}\frac{1}{k!}(y-x)^k\frac{\partial^k\psi(x)}{\partial y^k} \tag{4.27}$$

Substitute $a = y - x$ yields:

$$\psi(x+a) = \Big[\sum_{k=0}^{\infty} \frac{1}{k!}\Big(a\frac{\partial}{\partial x}\Big)^k\Big]\psi(x) \tag{4.28}$$

We then recognise the the sum is the exponential function, and write the wavefunctions in bra-ket notation.

$$\langle x+a|\psi\rangle = e^{a\frac{\partial}{\partial x}}\langle x|\psi\rangle \tag{4.29}$$

Generalising the 3 dimensions yields:

$$\langle \vec{x}+\vec{a}|\psi\rangle = e^{\vec{a}\cdot\nabla}\langle \vec{x}|\psi\rangle \tag{4.30}$$

Where we have defined Del operator to be:

$$\nabla \equiv \sum_i \hat{e}_i \frac{\partial}{\partial x_i} \tag{4.31}$$

To proceed, we consider the effect of translation operator on a definite state of position:

$$\hat{T}(a)|x\rangle = e^{-ia\hat{p}/\hbar}|x\rangle = |x+a\rangle \tag{4.32}$$

Take the Hermitian adjoints of both sides and right multiply by $|\psi\rangle$ yields:

$$\langle x|e^{ia\hat{p}/\hbar}|\psi\rangle = \langle x+a|\psi\rangle \tag{4.33}$$

Equate to (4.28) yields:

$$\langle x|e^{ia\hat{p}/\hbar}|\psi\rangle = e^{a\frac{\partial}{\partial x}}\langle x|\psi\rangle \tag{4.34}$$

Equate the terms that are first order in a in the Taylor series expansion of the exponential function yields:

$$\frac{ia}{\hbar}\langle x|\hat{p}|\psi\rangle = a\frac{\partial\langle x|\psi\rangle}{\partial x} \tag{4.35}$$

Cancelling a on both sides and multiply by $-i\hbar$ yields the final result:

$$\boxed{\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{\partial\langle x|\psi\rangle}{\partial x}} \tag{4.36}$$

The equation can also be generalised to 3 dimensions:

$$\langle \vec{x}|\hat{\vec{p}}|\psi\rangle = -i\hbar\nabla\langle \vec{x}|\psi\rangle \tag{4.37}$$

By analogy, provided there is no external torque, angular momentum is conserved under rotation. Thus we associate angular momentum with rotational symmetry and define angular momentum as the generator of rotation.

The rotation operator can thus be written as:

$$\boxed{\hat{R}(\vec{\alpha}) = e^{-\vec{\alpha}\cdot\hat{\vec{J}}/\hbar}} \tag{4.38}$$

Where $\vec{\alpha}$'s direction $\hat{\alpha}$ ($\hat{\cdot}$ now means unit vector instead of operator) be the axis of rotation and the length of the vector α corresponds to the angle of rotation.

$$\boxed{i\hbar\frac{\partial|\psi\rangle}{\partial\alpha} = \hat{\alpha}\cdot\hat{\vec{J}}|\psi\rangle} \tag{4.39}$$

When the angular momentum operator acts on a state, it gives the the rate of change at which the state changes as we rotate the system.

For small rotation $\delta\vec{\alpha}$, using 3D geometry yields:

$$\langle\hat{\vec{x}}\rangle' = \langle\hat{\vec{x}}\rangle + \delta\vec{\alpha}\times\langle\hat{\vec{x}}\rangle \tag{4.40}$$

$$\langle\psi|\hat{R}^\dagger(\delta\vec{\alpha})\hat{\vec{x}}\hat{R}(\delta\vec{\alpha})|\psi\rangle = \langle\psi|\hat{\vec{x}} + \delta\vec{\alpha}\times\hat{\vec{x}}|\psi\rangle \tag{4.41}$$

$$\Big(\hat{I} + \frac{i}{\hbar}\delta\vec{\alpha}\cdot\hat{\vec{J}}\Big)\hat{\vec{x}}\Big(\hat{I} - \frac{i}{\hbar}\delta\vec{\alpha}\cdot\hat{\vec{J}}\Big) = \hat{\vec{x}} + \delta\vec{\alpha}\times\hat{\vec{x}} \tag{4.42}$$

Equating the terms that are first order in $\delta\vec{\alpha}$ and only consider a single component $\hat{x}_i$ yields:

$$\frac{i}{\hbar}\delta\vec{\alpha}\cdot\hat{\vec{J}}\hat{x}_i - \frac{i}{\hbar}\hat{x}_i\delta\vec{\alpha}\cdot\hat{\vec{J}} = (\delta\vec{\alpha}\times\hat{\vec{x}})_i \tag{4.43}$$

$$\frac{i}{\hbar}[\delta\vec{\alpha}\cdot\hat{\vec{J}},\hat{x}_i] = (\delta\vec{\alpha}\times\hat{\vec{x}})_i \tag{4.44}$$

We now use the **Levi-Civita epsilon** ϵ_{ijk} to define the cross product between two vectors. It is recommended for the reader to expand the sum to check the definition is consistent with the result obtained by expanding determinant typically taught to students first.

$$\epsilon_{ijk} \equiv \begin{cases} +1, & \text{if } ijk = 123, 231, \text{or } 312, \\ -1, & \text{if } ijk = 321, 132, \text{or } 213, \\ 0, & \text{if } i=j, j=k, \text{or } k=i. \end{cases} \tag{4.45}$$

$$(\vec{v}\times\vec{w})_i \equiv \sum_{jk}\epsilon_{ijk}v_j w_k \tag{4.46}$$

We now use the summation notation to write the dot product and cross product, also using the rules for manipulating commutators to take the sum and number out of the commutator:

$$\frac{i}{\hbar}\sum_j \delta\alpha_j[\hat{J}_j,\hat{x}_i] = \sum_{jk}\epsilon_{ijk}\delta\alpha_j\hat{x}_k \tag{4.47}$$

Cancel the sum over j and $\delta\alpha_j$ on both sides and multiply by $-i\hbar$ yields:

$$[\hat{J}_j,\hat{x}_i] = -i\hbar\sum_k \epsilon_{ijk}\hat{x}_k \tag{4.48}$$

Using the fact that the Levi-Civita epsilon is anti-symmetric in the sense that $-\epsilon_{ijk} = \epsilon_{jik}$ and both i and j are dummy variables to obtain the important result.

$$[\hat{J}_j, \hat{x}_i] = i\hbar \sum_k \epsilon_{jik} \hat{x}_k \tag{4.49}$$

$$[\hat{J}_i, \hat{x}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{x}_k \tag{4.50}$$

This result can be generalised by replacing $\hat{x}_i$ with any vector operator as we have only used the fact that the operator is a vector when deriving the commutation relation. For example, if we rotate the entire system, the momentum vector is also rotated. One of the most important commutation relation is obtained by replacing $\hat{x}_j$ with $\hat{J}_j$ itself:

$$\boxed{[\hat{J}_i, \hat{J}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{J}_k} \tag{4.51}$$

This has the far reaching consequence: It is not possible to simultaneously and precisely specify the angular momentum in two orthogonal directions. Quantitatively, substitute (4.51) into the uncertainty principle (3.22) yields:

$$\sigma_{J_x} \sigma_{J_y} \geq \frac{\hbar}{2} |\langle \hat{J}_z \rangle| \tag{4.52}$$

Therefore, the more angular momentum there is in one direction, the greater the minimum uncertainties in the two orthogonal directions.

To derive the commutation relation between angular momentum operator and scalar operator, we start by observing that the expectation of a scalar operator $\hat{S}$ is invariant under rotation.

$$\langle \psi' | \hat{S} | \psi' \rangle = \langle \psi | \hat{S} | \psi \rangle \tag{4.53}$$

$$\hat{R}^\dagger \hat{S} \hat{R} = \hat{S} \tag{4.54}$$

We now left multiply both sides by $\hat{R}$, noting it is unitary:

$$\hat{S}\hat{R} = \hat{R}\hat{S} \tag{4.55}$$

Hence, all scalar operators commute with rotation operator, as rotation operator is a function of angular momentum operator, we can conclude:

$$[\hat{J}_i, \hat{S}] = 0 \tag{4.56}$$

The angular momentum operator commutes with all scalar operators, one important scalar operator is the total angular momentum squared operator $\hat{J}^2 = \sum_i \hat{J}_i^2$, which has the following commutation relation:

$$\boxed{[\hat{J}_i, \hat{J}^2] = 0} \tag{4.57}$$

Hence, there exist a complete set of mutual eigenstates of both operators. It is possible to simultaneously and precisely specify both the total angular momentum and the angular momentum along only one axis by choice.

We have found that momentum is the generator of translation, angular momentum is the generator of rotation. But the remaining question is: What generates time evolution?

In an isolated system, energy is always conserved, furthermore, physics laws does not change from day to day. Thus we associate energy conservation with time evolution symmetry.

Energy is the generator of time evolution.

5 Time evolution

At the end of the last chapter, we obtained a profound result that energy is the generator of time evolution, thus we discovered how to evolve our quantum state in time. The Schrödinger equation (Time-dependent) or **TDSE** is exactly this statement.

$$\boxed{i\hbar\frac{\partial|\psi\rangle}{\partial t} = \hat{H}|\psi\rangle} \tag{5.1}$$

We have arrived at the fifth and the last postulate of quantum mechanics:
Quantum state evolves in time in accordance with TDSE
It is important to note that this equation works even when the Hamiltonian is time dependent. The time evolution operator is:

$$\hat{U}(t) = e^{-it\hat{H}/\hbar} \tag{5.2}$$

Hamiltonian is time-independent in the time evolution operator. TDSE is the equation of motion in quantum mechanics it is the conceptual equivalent of Newton's second law in classical mechanics. However, there are a few crucial differences. Firstly, TDSE is first order differential equation whereas Newton's second law is second order differential equation as acceleration is the second order time derivative of position. Thus to solve Newton's second law, both the initial position and velocity is required. However, to solve TDSE, only the initial state of the system is required, which is fully encoded in the ket $|\psi\rangle$. This is consistent with our initial construction of the $|\psi\rangle$ as a mathematical entity that represents full knowledge of the system. The solution of Newton's second law is position as a function of time $\vec{x}(t)$ also known as a trajectory. The solution of TDSE is the state of the system at some future time $|\psi(t)\rangle$. We now begin to solve the TDSE.

We begin by substituting the eigenstates of Hamiltonian $|n\rangle$ for $|\psi\rangle$.

$$i\hbar\frac{\partial|n\rangle}{\partial t} = \hat{H}|n\rangle = E_n|n\rangle \tag{5.3}$$

The first order differential equation has a rather simple solution:

$$|n,t\rangle = e^{-iE_nt/\hbar}|n\rangle \tag{5.4}$$

This shows that the time evolution of energy eigenstate is simply a rotation in the complex plane, the rate of rotation is proportional to energy of the state. We now expand a generic state in the basis of energy eigenstates at some arbitrary moment in time, as energy eigenstates form a complete set at all time.

$$|\psi(t)\rangle = \sum_n a_n(t)|n,t\rangle \tag{5.5}$$

Substitute into the TDSE yields:

$$\sum_n (i\hbar\dot{a}_n + a_nE_n)e^{-iE_nt/\hbar}|n\rangle = \sum_n a_nE_ne^{-iE_nt/\hbar}|n\rangle \tag{5.6}$$

Where we have differentiated with respect to time and multiplied by $i\hbar$ on LHS and applied the Hamiltonian operator on RHS. Cancellation of common terms on both sides of (5.6) yields:

$$\dot{a}_n = 0 \tag{5.7}$$

We have obtained the result that the amplitude in energy representation is time independent, thus remains constant in time. We have also solved the TDSE, the state at a future time t can be written as:

$$\boxed{|\psi(t)\rangle = \sum_n a_n e^{-iE_n t/\hbar}|n\rangle} \tag{5.8}$$

where $a_n = \langle n|\psi(0)\rangle$

Hence, if we solve the TISE for a system, we can evolve any arbitrary quantum state in time and understand its dynamics. This is why a lot of quantum mechanics is centred around solving the TISE.

We are now in a position to demonstrate quantum mechanics is consistent with classical mechanics. We will first consider the time evolution of the expectation value of an observable, as we have already established at the end of chapter 2 that classical mechanics is the mechanics of expectation values where the quantum fluctuation around the average value is negligible.

The Hermitian adjoint of TDSE is:

$$-i\hbar\frac{\partial\langle\psi|}{\partial t} = \langle\psi|\hat{H} \tag{5.9}$$

Where we have used the fact that the Hamiltonian is a Hermitian operator.

$$i\hbar\frac{d\langle\hat{Q}\rangle}{dt} = i\hbar\frac{d\langle\psi|\hat{Q}|\psi\rangle}{dt} \tag{5.10}$$

$$i\hbar\frac{d\langle\hat{Q}\rangle}{dt} = i\hbar\frac{\partial\langle\psi|}{\partial t}\hat{Q}|\psi\rangle + i\hbar\langle\psi|\frac{\partial\hat{Q}}{\partial t}|\psi\rangle + i\hbar\langle\psi|\hat{Q}\frac{\partial|\psi\rangle}{\partial t} \tag{5.11}$$

Substituting (5.1) and (5.9) in (5.11) yields:

$$i\hbar\frac{d\langle\hat{Q}\rangle}{dt} = -\langle\psi|\hat{H}\hat{Q}|\psi\rangle + i\hbar\langle\psi|\frac{\partial\hat{Q}}{\partial t}|\psi\rangle + \langle\psi|\hat{Q}\hat{H}|\psi\rangle \tag{5.12}$$

Using the definition of commutator and the expression for expectation value yields:

$$i\hbar\frac{d\langle\hat{Q}\rangle}{dt} = \langle[\hat{Q},\hat{H}]\rangle + i\hbar\Big\langle\frac{\partial\hat{Q}}{\partial t}\Big\rangle \tag{5.13}$$

Dividing both sides by $i\hbar$ yields the **Ehrenfest theorem**, thus we made our connection back to classical mechanics.

$$\boxed{\frac{d\langle\hat{Q}\rangle}{dt} = \frac{1}{i\hbar}\langle[\hat{Q},\hat{H}]\rangle + \Big\langle\frac{\partial\hat{Q}}{\partial t}\Big\rangle} \tag{5.14}$$

We now justify our choices for generators in chapter 4. We assume the operator representing the observable of interest is time-independent, this is usually true as the spectrum does not change in time. The Hamiltonian governs the dynamics of the system as it tells the system how to move forward in time. If the dynamics is invariant under a

symmetry operation, then the Hamiltonian commutes with the symmetry operation and the generator that transformation. For translational symmetry:

$$[\hat{p}, \hat{H}] = 0 \tag{5.15}$$

Using Ehrenfest theorem:

$$\frac{d\langle\hat{p}\rangle}{dt} = 0 \tag{5.16}$$

This demonstrates it is the right choice to associate conservation of momentum with translational symmetry and defining momentum as the generator of translation.

Similarly, for a system that is invariant under rotation:

$$[\hat{\vec{J}}, \hat{H}] = 0 \tag{5.17}$$

Using Ehrenfest theorem:

$$\frac{d\langle\hat{\vec{J}}\rangle}{dt} = 0 \tag{5.18}$$

This demonstrates it is the right choice to associate conservation of angular momentum with rotational symmetry and defining angular momentum as the generator of rotation.

Lastly, the Hamiltonian commutes with itself.

$$[\hat{H}, \hat{H}] = 0 \tag{5.19}$$

Using Ehrenfest theorem:

$$\frac{d\langle\hat{H}\rangle}{dt} = 0 \tag{5.20}$$

This is another way of phrasing the the law of conservation of energy in an isolated system, but also the dynamics of system stays the same as we move the system in time.

This demonstrates it is the right choice to associate conservation of energy with time evolution symmetry and justifies our choice of energy as the generator of time evolution.

For operators that commutes with the Hamiltonian, then the observable is called **constant of motion** thus a conserved quantity and the eigenvalues of the operator is said to be a **good quantum number** and is usually used to distinguish the different states with the same energy.

We now attempt to gain more insight into these energy eigenstates $|n\rangle$ by substituting them into (5.12), assume $\hat{Q}$ is time-independent:

$$i\hbar\frac{d\langle\hat{Q}\rangle}{dt} = -\langle n|\hat{H}\hat{Q}|n\rangle + \langle n|\hat{Q}\hat{H}|n\rangle \tag{5.21}$$

Using TISE and the fact that Hamiltonian is Hermitian to operate back into $\langle n|$ yields:

$$i\hbar\frac{d\langle\hat{Q}\rangle}{dt} = -E_n\langle n|\hat{Q}|n\rangle + E_n\langle n|\hat{Q}|n\rangle \tag{5.22}$$

$$\frac{d\langle\hat{Q}\rangle}{dt} = 0 \tag{5.23}$$

Hence, if the state is an energy eigenstate, then the expectation value of all time-independent operator is constant in time. Another way to look at this is that energy

eigenstates are not localised in time analogous to momentum eigenstates can be anywhere in space. We shall also call energy eigenstates **stationary states** to emphasis this point. Stationary states are again an idealised mathematical construct, but they have a crucial role with constructing the solution of TDSE as in (5.8), so that we can easily evolve any realistic state in time.

We now use the Ehrenfest theorem to evaluate the rate of change of expectation value of position and momentum to gain more insight into the connection between the quantum mechanics and classical mechanics. Both position and momentum operators are time-independent as the spectrum of both physical quantities remain continuous and is unchanged in time.

The form of the Hamiltonian is constructed by analogy to the classical expression: Kinetic + Potential energy.

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \tag{5.24}$$

We first evaluate the rate of change of position by evaluating the following commutator.

$$[\hat{x}, \hat{H}] = \left[\hat{x}, \frac{\hat{p}^2}{2m} + V(\hat{x})\right] \tag{5.25}$$

Expand the commutator yields:

$$[\hat{x}, \hat{H}] = \left[\hat{x}, \frac{\hat{p}^2}{2m}\right] + [\hat{x}, V(\hat{x})] \tag{5.26}$$

Taking out the constant in the first commutator on RHS and noting any operator commutes with functions of itself yields:

$$[\hat{x}, \hat{H}] = \frac{1}{2m}[\hat{x}, \hat{p}^2] \tag{5.27}$$

Using (3.3) yields:

$$[\hat{x}, \hat{H}] = \frac{2\hat{p}}{2m}[\hat{x}, \hat{p}] \tag{5.28}$$

Using the canonical commutation relation yields:

$$[\hat{x}, \hat{H}] = i\hbar\frac{\hat{p}}{m} \tag{5.29}$$

Finally, substituting back into Ehrenfest theorem (5.14) and rearrange to obtain:

$$\boxed{\langle\hat{p}\rangle = m\frac{d\langle\hat{x}\rangle}{dt}} \tag{5.30}$$

This is the definition of momentum in classical mechanics, hence our definition of momentum as the generator of translation is consistent with the classical definition.

We now carry out the same calculation for the rate of change of momentum.

$$[\hat{p}, \hat{H}] = \left[\hat{p}, \frac{\hat{p}^2}{2m} + V(\hat{x})\right] = [\hat{p}, V(\hat{x})] \tag{5.31}$$

$$= [\hat{p}, \hat{x}]\frac{d\hat{V}}{dx} = -i\hbar\frac{d\hat{V}}{dx} \tag{5.32}$$

Substitute back into (5.14) yields the final result:

$$\frac{d\langle\hat{p}\rangle}{dt} = \left\langle -\frac{d\hat{V}}{dx} \right\rangle \tag{5.33}$$

The RHS of (5.33) is the negative gradient of potential energy, we recognise this quantity as the classical definition of force. With force operator conveniently defined as $\hat{F} \equiv -d\hat{V}/dx$, we obtain the final result:

$$\boxed{\langle\hat{F}\rangle = \frac{d\langle\hat{p}\rangle}{dt}} \tag{5.34}$$

Hence, we have recovered Newton's second law. This demonstrates quantum mechanics is consistent with classical mechanics for large particles. This also demonstrates the form of the Hamiltonian as a sum of kinetic and potential energy is correct. Classical mechanics can therefore be regarded as a very good approximation of quantum mechanics at macroscopic scale.

Lastly, let us look at how quantum mechanics describes the dynamics of a realistic free particle.

A free particle is not bound by a potential thus the Hamiltonian of the system is given by:

$$\hat{H} = \frac{\hat{p}^2}{2m} \tag{5.35}$$

Thus momentum eigenstates $|p\rangle$ are also energy eigenstates and the solutions to TISE. We can immediately construct the solution of TDSE, by generalising equation (5.8) to the continuous analogue.

$$|\psi(t)\rangle = \int dp\, \langle p|\psi(0)\rangle e^{-ip^2t/2m\hbar}|p\rangle \tag{5.36}$$

Suppose the momentum of the particle is normally distributed around some mean momentum p_0 with σ as the standard deviation/uncertainty in initial position. The wavefunction is given by equation (3.33) with a small modification.

$$\langle p|\psi(0)\rangle = \frac{1}{(2\pi\hbar^2/4\sigma^2)^{1/4}} e^{-(p-p_0)^2\sigma^2/\hbar^2} \tag{5.37}$$

We then left multiply both sides of (5.36) by $\langle x|$ to go into position representation, as we wish to see what is happening in physical space. Substituting in (5.37) and (3.27) yields:

$$\langle x|\psi(t)\rangle = \frac{1}{(2\pi\hbar^2/4\sigma^2)^{1/4}\sqrt{h}} \int dp\, e^{-(p-p_0)^2\sigma^2/\hbar^2} e^{-ip^2t/2m\hbar} e^{ipx/\hbar} \tag{5.38}$$

After some tedious algebra, which is left as an exercise for the reader, we obtain:

$$\langle x|\psi(t)\rangle = \frac{e^{ip_0^2t/2m\hbar}\sqrt{\pi}}{(2\pi\hbar^2/4\sigma^2)^{1/4}\sqrt{h}\xi} e^{ip_0(x-p_0t/m)/\hbar} e^{-(x-p_0t/m)^2/4\hbar^2\xi^2} \tag{5.39}$$

Where:

$$\xi^2 = \frac{\sigma^2}{\hbar^2} + \frac{it}{2m\hbar} \tag{5.40}$$

The result above is quite complicated, the probability distribution is much simpler:

$$|\langle x|\psi(t)\rangle|^2 = \frac{\sigma}{\sqrt{2\pi}\hbar^2|\xi|^2} e^{-(x-p_0t/m)^2\sigma^2/2\hbar^4|\xi|^4} \tag{5.41}$$

This is a normal distribution with a time-dependent mean $\mu(t)$ and variance $\sigma^2(t)$ of:

$$\mu(t) = \frac{p_0 t}{m} \tag{5.42}$$

$$\sigma^2(t) = \sigma^2 + \left(\frac{\hbar t}{2m\sigma}\right)^2 \tag{5.43}$$

Hence, the particle is travelling at a mean velocity of p_0/m just as expected. To make sense of the increasing in variance as the particle evolves in time we note the initial uncertainty in momentum is $\sigma_p = \hbar/2\sigma$ from the end of chapter 3, this gives the extra uncertainty of $\hbar t/2m\sigma$ in position over time. This can be understood as due to the initial uncertainty in momentum, hence velocity, particles that are faster initially is further ahead and particles that are slower initially is lagged behind at time t. This is a perfectly sensible physical picture.

To conclude, this demonstrates quantum mechanics is able to describe a realistic behaviour of a free particle. However, the way it reaches this conclusion is rather exceptional. We began by writing a realistic state as a quantum superposition of definite states of momentum and energy $|p\rangle$ which are neither localised in space or time. We then evolved these unphysical states forward in time and constructed the final state as another superposition of these unphysical states. Lastly, we took the modulus square of the wavefunction and obtained something that is correct based on our mere physical intuition.

Quantum mechanics is the most accurate and complete theory of the physical reality, however, it is amazing that no one really understands why the theory works and many fundamental questions of the theory is left unanswered up to today.